

# Tutorial

1. Define the term malware.
2. State one difference between a virus and a worm.
3. Explain one limitation of a firewall.
4. Explain the purpose of encryption.
5. State one advantage of asymmetric encryption over symmetric encryption.
6. Explain the purpose of a digital signature.
7. State one authentication method other than passwords.
8. Compare symmetric encryption and asymmetric encryption.
9. Compare how a firewall and an intrusion prevention system (IPS) control network traffic.
10. Compare encryption and digital signatures in ensuring network security.

11. A school wants to improve the security of its online learning system. Describe how the following could be used to improve security:

- firewall,
- IDS or IPS,
- encryption,
- authentication.

For each method, explain one limitation.

[8]

1. **Firewall:** Blocks unauthorized external traffic by checking packet headers (IPs and ports).
2. **IDS/IPS:** Find and block malware signatures in payloads
3. **Encryption:** Scrambles data to make intercepted data unreadable
4. **Authentication:** Multi-factor authentication to allow verified access

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- IDS or IPS,
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- authentication.

For each method, explain one limitation.

[8]

1. **Firewall:** cannot detect insider threats
2. **IDS/IPS:** false positives/ false negatives
3. **Encryption:** possible to hide encrypted malware pass firewall
4. **Authentication:** insider threats cannot be prevented

12. Customers of a bank have to use the Internet in order to do electronic banking. Explain the problem of maintaining confidentiality of data on the Internet and the need for techniques to address the problem. [5]

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### **Problem of maintaining confidentiality**

- Hackers attack communication/ customers' data
- Data being distributed leading to unsolicited communications

### **Techniques to address problem**

- Encrypting data on database files
- Digital signatures to guarantee reliability of source
- Passwords to enter user's area/database
- Use of firewall to block unwanted access
- Portable storage not allowed
- Use of authorisation and authentication techniques

13. Explain the difference between an Intrusion Detection System (IDS) and an Intrusion Prevention System (IPS). Explain how each system improves network security and state one limitation of each. [6]

### Detection systems:

- **Signature-based** detections perform simple pattern-matching of a known attack
- **Heuristic** detections, a.k.a. **anomaly based** ones, learn what is normal for a network and flag any deviations from this baseline as potential threats
  - For e.g. Sarah who works at a company typically logs into the company's network every weekday morning at 9:00 AM from her office computer. This behavior is considered normal or baseline for Sarah. However, one day, the system detects a deviation from this baseline.
  - One day, the system notices that Sarah logs into the network at an unusual time, such as 2:00 AM, from a different location, using a different device (e.g., a mobile phone or a laptop). Additionally, the system observes that Sarah is accessing sensitive files or areas of the network that she doesn't typically access.

| IDS                                                                                                                                                                                                                                                                                                                                                                                                                                         | IPS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Both are security mechanisms used to detect and respond to suspicious or malicious activities within a network. However, they differ in approach and level of action taken.</li> <li>Choice depends on organisation's security requirements, resources, and risk tolerance.</li> <li>A layered defense strategy of implementing both provides comprehensive approach to network security.</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <ul style="list-style-type: none"> <li>monitors network traffic</li> <li>analyses traffic for signs of potential security breaches or attacks</li> <li>generates alerts</li> <li>provides information about potential threats</li> <li>For example, IDS detects sudden surge of network traffic from an external IP address, may generate alert indicating a possible DDoS attack.</li> </ul>                                               | <ul style="list-style-type: none"> <li>not only detects malicious activities but also takes immediate action to mitigate or stop them</li> <li>can dynamically modify network traffic, block specific IP addresses, or apply access control rules to prevent attacks from reaching their intended targets</li> <li>For example, IPS identifies a known exploit attempting to target a vulnerable service, can automatically drop or modify network packets to block the attack.</li> </ul> |

14. Explain how firewalls restrict unauthorised access to networks and state two limitations of firewalls. [4]

- helps protect networks from unauthorized access and malicious activities,
- acts as a barrier between internal and external networks,
- monitors and controls network traffic based on predefined security rules,
- filtering rule most commonly based on IP addresses, port numbers, protocols
- other filtering on content of packet (keywords, patterns, file types)



15. As an employee at a financial institution, you received a loan application from a client electronically.

(a) Discuss how digital signatures are used to secure and validate the loan application process. [4]

(b) Explain the advantages of using digital signatures in this context, both for the financial institution and the client. [4]

(a) The financial institution employs digital signatures to secure and validate the loan application process. When a client submits a loan application electronically, the application is digitally signed to ensure its integrity and authenticity.

Digital signatures binds the identity of the signer to the document or transaction, providing assurance of its origin and integrity.

15. As an employee at a financial institution, you received a loan application from a client electronically.

(a) Discuss how digital signatures are used to secure and validate the loan application process. [4]

(b) Explain the advantages of using digital signatures in this context, both for the financial institution and the client. [4]

Steps:

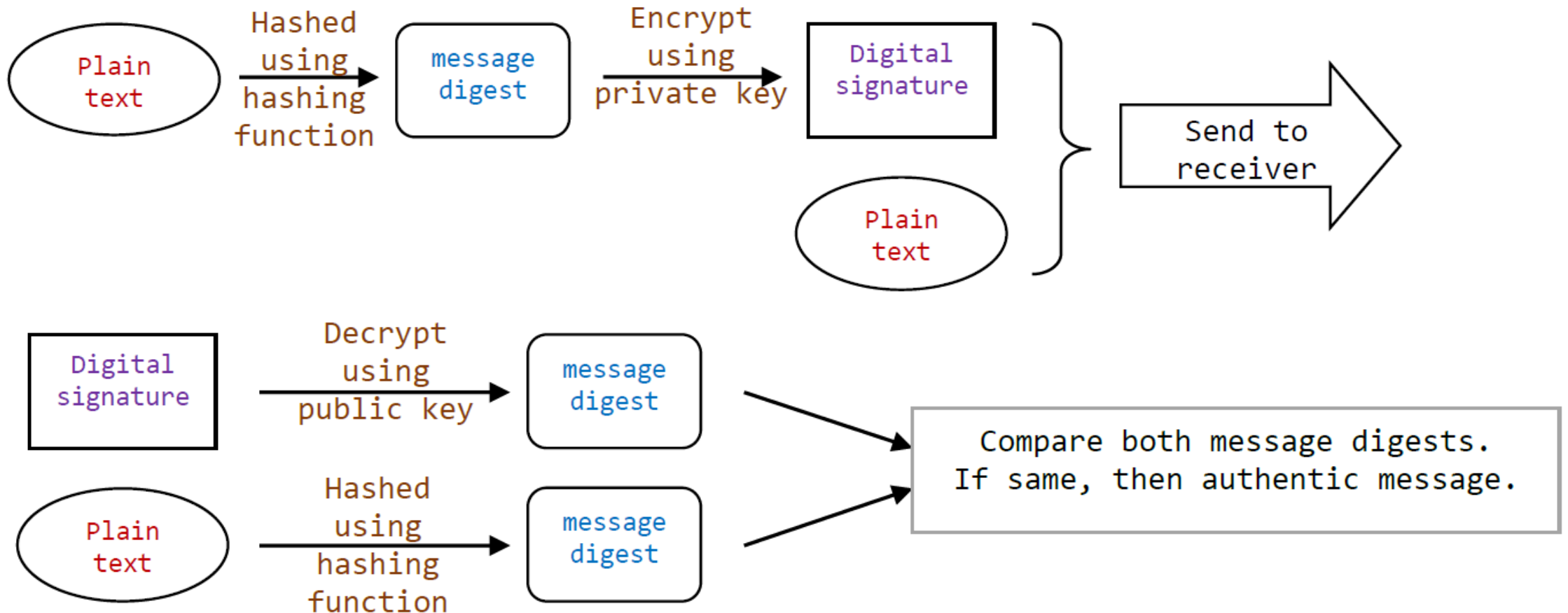
1) Loan application document is hashed (using a mathematical function), creating a unique digital fingerprint (message digest) that represents its content.

2) Client's private key is used to encrypt the message digest (into digital signature). This encrypted hash, along with the client's loan application is sent to the financial institution.

3) When financial institution receives the digitally signed loan application, it will verify its authenticity and integrity using the client's public key. The digital signature is thus decrypted. The hash of the received document is also calculated using the same hash algorithm as the sender.

4) By comparing this calculated hash with the decrypted hash from the digital signature, financial institution can ensure that the loan application has not been tampered with or altered since it was signed.

# Digital Signature



15. As an employee at a financial institution, you received a loan application from a client electronically.

(a) Discuss how digital signatures are used to secure and validate the loan application process. [4]

(b) Explain the advantages of using digital signatures in this context, both for the financial institution and the client. [4]

1) Provide strong non-repudiation, meaning client cannot deny their involvement or authorisation of the loan application. This offers legal protection to the financial institution in case of disputes.

2) Ensure the integrity of the loan application. Any modification or tampering with the document would result in a failed signature verification, alerting the financial institution to potential fraud attempts or unauthorised changes.

3) Streamline the loan application process by eliminating the need for physical paperwork and manual verification. This reduces administrative overhead and enhances efficiency, allowing for faster processing times and improved customer experience.

4) Offer enhanced security compared to traditional paper-based signatures. The private keys used for signing are protected by strong encryption and can be stored securely, reducing the risk of unauthorised access.

## 16. 9569/01/SP/2020

A company has many offices at different sites in many other countries. Each office is able to access data stored on the network servers at the company headquarters. The offices access data using the Internet.

- (a) Describe two threats and how they could compromise company servers. For each threat, explain how the servers and data stored on them could be protected and identify any limitations of the protection. [8]
- (b) Data transferred over the internet from the company headquarters needs to be kept secure. Explain three security features that should be used in order to ensure this data would be protected before data transfer. [6]

(a)

- **Malware** can be introduced as a worm, virus, etc. Software that deliberately causes harm to data or software held on the computer servers. Install and update antivirus software. New malware may not be recognised.
- **Distributed denial of service** attacks that flood the servers with requests so that the system becomes overloaded and the servers are no longer available to company employees. Use firewall to filter suspicious traffic. Large-scale attacks may still overwhelm system.

**(b) Focus is on protecting data during transmission over the Internet.**

- **Encryption** can convert data into an unreadable format using cryptographic algorithms. Need decryption key to transform data back into original readable form. Intercepted data cannot be read without key.
- **Authentication.** Usernames, passwords and multi-factor authentication should verify that only authorized users can access servers.
- A **Virtual Private Network** is a secure and private connection over Internet. Data sent through **encrypted channel** through remote server that ensure that sensitive data is safely transmitted. Prevents unauthorised people from eavesdropping on the traffic and allows the user to conduct work remotely.

17. Explain why a company should use multiple security measures such as firewalls, encryption, antivirus software, and authentication instead of relying on only one method.

[4]

- **Remove single point of failure** with different methods
- Different tools tackle different threats: **firewall** blocks outside network intrusion, **antivirus** removes malicious files on computer
- When file stolen, **encryption** ensures file remains unreadable
- Ensure verified users gain access only using **authentication**



18. Explain how malware or a denial of service (DoS) attack could compromise a computer system or network. State one method that could reduce the risk of each attack. [4]

## Malware

- **Executes malicious code** to steal data, corrupt files, or encrypt systems for ransom.
- Deploy and continuously update **antivirus software** to scan and isolate malicious payloads.

## Denial of service

- **Floods servers** with fake traffic, exhausting bandwidth and CPU to crash services for legitimate users.
- **Configure firewalls** to block sudden traffic spikes and drop excessive requests.